

PROJECT 1

CLOUD READINESS & MIGRATION ASSESSMENT

Topic:

Cloud Readiness Assessment for an E-Commerce Web Application

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TABLE OF CONTENTS

Sr. No.	Contents	Page No.
1	Introduction	1
2	Problem Statement	2
3	Existing Infrastructure Overview	4
4	Current Challenges & Existing System Issues	6
5	Cloud Readiness Assessment	8
6	Gap Analysis	11
7	Application Dependency Mapping	13
8	Proposed AWS Cloud Architecture	16
9	TCO vs ROI Comparison	19
10	Cloud Maturity Score & Assessment Matrix	22
11	Migration Plan & Migration Strategy	25
12	Security Recommendations & Best Practices	29
13	Benefits of Cloud Migration	32
14	Conclusion	34

INTRODUCTION

Cloud computing is really important for companies these days. Many organizations are switching from systems to cloud platforms because they want to be able to handle more work be available all the time be secure and run smoothly.

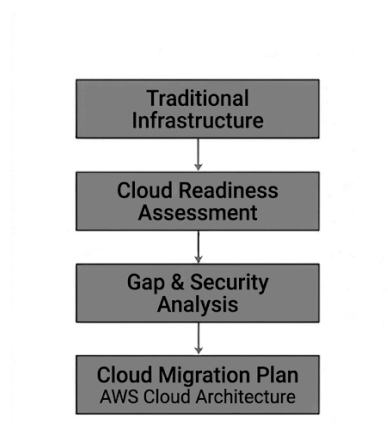
The old way of doing things is expensive to maintain. You have to take care of the hardware yourself do backups by hand and have a complicated plan in place in case something goes wrong. This can hurt the business. Make the applications run slowly when a lot of people are using them.

Cloud platforms are a way to do things because they are flexible and do not cost as much. They help organizations update their applications and infrastructure. You can get things like servers, databases that are taken care of for you, automatic backups, tools to monitor everything and storage that can grow with your needs. All of this helps make sure the applications are reliable and the business can keep running.

This project is about figuring out if an existing E-Commerce Web Application is ready to move to the cloud. The application is currently running on systems. We will look at the infrastructure check the security map out the dependencies compare the costs analyze how ready the company is for the cloud and make recommendations, for moving to the cloud.

We will also suggest an architecture that is secure and can grow with the companys needs using AWS cloud services. This will help the application run better be more efficient and be more secure. Cloud computing will help the E-Commerce Web Application be more scalable and run smoothly.

Cloud Migration Process :



PROBLEM STATEMENT

The company has its E-Commerce Web Application on its servers right now. People use the application to look at products register manage orders make payments and keep track of inventory.

The current system has a lot of problems that make it hard for the application to work properly.

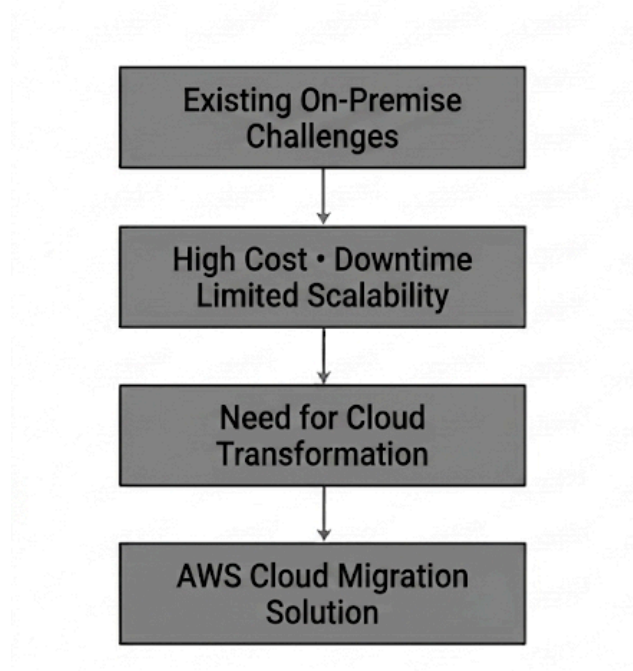
The big issues, with the system are:

- * The cost of taking care of the infrastructure is really high
- * When a lot of people are using the application at the time it gets slow
- * Backing up and recovering data has to be done by hand
- * The application is down when we are doing maintenance
- * We do not have a way to monitor everything and get alerts
- * The old systems have security problems
- * It is hard to get everything up and running if something bad happens
- * When a lot of people are using the application it gets slow

The company wants to move the E-Commerce Web Application to the AWS cloud platform. This will help the application work be more secure and cost less to run.

This project checks if the E-Commerce Web Application is ready to move to the cloud. It looks for problems, security risks and things that might make the move hard. The project also finds the best cloud solutions to make the move successful.

Infrastructure Transformation Process :



EXISTING INFRASTRUCTURE OVERVIEW

The E-Commerce Web Application is currently running on the organizations infrastructure, which is located in their local data center. This infrastructure is made up of servers that do several things, including hosting the application managing the database storing files and handling user authentication.

The application is built in a way where all the services are closely connected and are hosted on just a few servers. This way of deploying the application does not have automation, a way of monitoring or the ability to scale up easily.

The parts that make up the system are:

- * Frontend Web Application
- * Backend Application Server
- * MySQL Database Server
- * Local File Storage
- * Manual Authentication System
- * Local Monitoring System

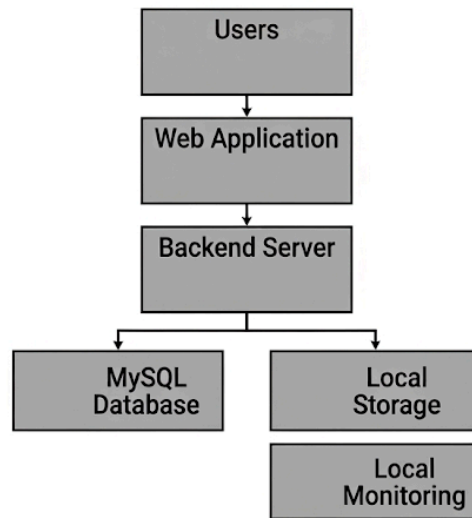
The current infrastructure needs upkeep of the hardware manual updates of the software and regular backups all of which have to be done by the IT support team.

Existing Infrastructure Table

Component	Description
Frontend Web Application	Customer-facing website
Backend Application Server	Handles business logic and APIs
Database Server	Stores customer and order data
File Storage	Stores images and application files

Authentication System	Manages user login and access
Monitoring System	Performs basic local monitoring

Existing On-Premise Architecture Diagram



CURRENT CHALLENGES AND EXISTING SYSTEM ISSUES

The E-Commerce application is having a lot of problems with the system we have in place. We are talking about the E-Commerce application and its infrastructure. The E-Commerce application is facing issues that are related to how it works the technology used and how secure it is. All these issues are affecting how well the E-Commerce application works and how reliable it is.

When more people use the E-Commerce application and do things online the system we have now is not able to handle all the traffic. The E-Commerce application is having trouble keeping up with the demand. Our organization is also spending a lot of money to keep the hardware working and to manage the infrastructure manually. This is costing the organization a lot of money.

The big problems, with the system used by the E-Commerce application are listed below.

Technical Challenges

Challenge	Description
Limited Scalability	Infrastructure cannot automatically handle increased traffic
Single Point of Failure	Failure of one server can affect the entire application
Manual Deployment	Software deployment process is slow and error-prone
Lack of Automation	Infrastructure provisioning and backups are manually managed
Performance Issues	Application response time increases during peak load
Limited Disaster Recovery	Recovery process is slow during system failures

Business Challenges

Challenge	Business Impact
High Maintenance Cost	Increased operational expenditure
Downtime During Maintenance	Affects customer experience and sales
Slow Feature Deployment	Delays business improvements
Infrastructure Expansion Difficulty	Limits future business growth

Security Challenges

Security Issue	Risk
Weak Access Control	Unauthorized access possibility
No Multi-Factor Authentication	Increased authentication risk
Limited Monitoring	Delayed incident detection
Manual Backup Management	Risk of data loss
No Data Encryption	Sensitive customer data exposure

CLOUD READINESS ASSESSMENT

Cloud readiness assessment is the process of evaluating whether an existing application and infrastructure are suitable for migration to the cloud environment.

The assessment helps identify technical limitations, security gaps, infrastructure compatibility, operational challenges, and business requirements before cloud migration.

The current E-Commerce Web Application was analyzed based on multiple assessment areas including infrastructure readiness, application compatibility, database readiness, scalability, monitoring capabilities, and security preparedness.

The assessment results indicate that the application is moderately ready for cloud migration with some required improvements in automation, monitoring, backup management, and security implementation.

Cloud Readiness Assessment Table

Assessment Area	Status	Remarks
Infrastructure Readiness	Medium	Existing servers can be migrated with minor modifications
Application Compatibility	High	Application supports cloud deployment
Database Readiness	Medium	Database optimization required before migration
Security Readiness	Low	Additional security controls required
Monitoring Readiness	Low	Centralized monitoring tools not implemented
Backup & Recovery	Medium	Backup process exists but is manual
Scalability Readiness	Medium	Limited support for automatic scaling

Readiness Classification

Readiness Level	Description
High	Fully prepared for migration
Medium	Migration possible with improvements
Low	Significant changes required before migration

Assessment Summary

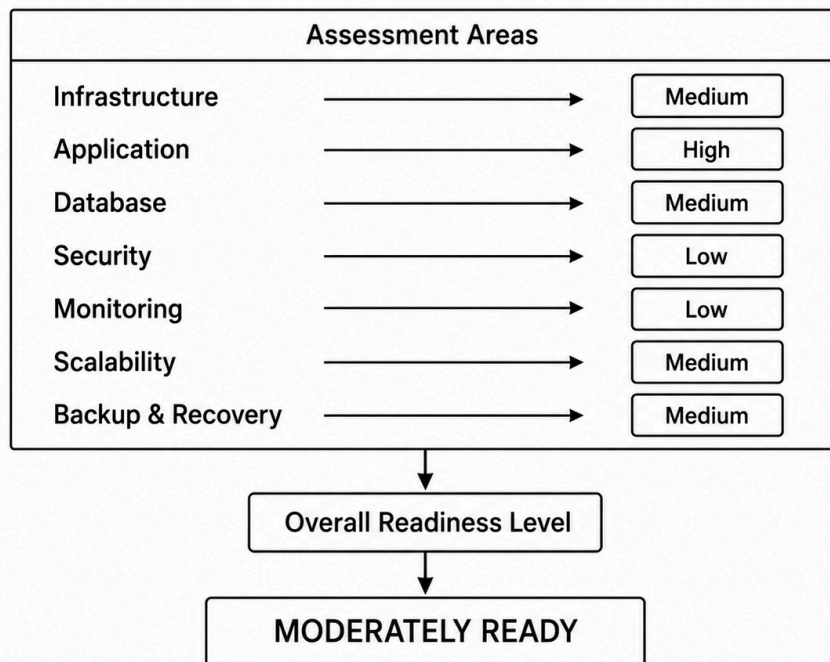
The assessment showed that our current setup is partly good for moving to the cloud. The way our application is built and our database works can be moved to Amazon Web Services with some changes.

We need to work on a few things:

- Automating our infrastructure
- Making security better
- Keeping an eye on things and alerting us when needed
- Managing backups
- Planning for disasters

If we move step by step and plan it well we can successfully move our application to the cloud.

CLOUD READINESS ASSESSMENT



GAP ANALYSIS

We do something called gap analysis to see what is missing between what our infrastructure can do and what it needs to do to meet the standards, for the cloud.

The analysis helps us find the problems, business issues, operational things that do not work well and security issues that we need to fix before we can move to the cloud.

Our current E-Commerce infrastructure has a lot of gaps when it comes to being able to handle work automating tasks keeping things secure watching what is going on and recovering from disasters.

Technical Gap Analysis

Current State	Gap Identified	Recommended Improvement
Monolithic Architecture	Limited scalability and flexibility	Gradual migration towards modular architecture
Manual Deployment Process	Increased deployment time and human error	CI/CD pipeline implementation
Local Storage System	Limited storage scalability	Cloud object storage solution
Single Server Dependency	High risk of service failure	Load-balanced multi-instance deployment
Manual Backup Process	Risk of backup failure and data loss	Automated cloud backup services
Limited Monitoring	Delayed issue detection	Centralized cloud monitoring tools

Business Gap Analysis

Current Business Issue	Impact on Organization
High Infrastructure Cost	Increased operational expenditure
Downtime During Maintenance	Reduced customer satisfaction
Slow Feature Release	Delayed business improvements
Scalability Limitations	Difficulty handling future growth
Resource Management Complexity	Increased administrative overhead

Security Gap Analysis

Existing Security Gap	Recommended Solution
Weak Access Control	Implement IAM-based role management
No Multi-Factor Authentication	Enable MFA for users and administrators
Limited Log Monitoring	Centralized logging using CloudWatch
No Encryption Mechanism	Encrypt data at rest and in transit
Manual Security Updates	Automated patch management
Limited Backup Security	Secure cloud-based backup policies

Gap Analysis Summary

The gap analysis showed we need to fix things before moving everything to the cloud.

The main issues are, with automating our infrastructure monitoring everything from one place putting in place security measures managing how scalable our systems are, and planning for disaster recovery.

Using AWS cloud services and modern cloud practices can really help us improve how reliable our applications are make our security better make our operations more efficient and ensure our business can keep going even if something goes wrong.

APPLICATION DEPENDENCY MAPPING

Application dependency mapping is about finding out how different parts of an application are connected. These parts include services, databases, storage systems and external integrations.

Before moving to the cloud it's crucial to analyze dependencies. This helps us see how modules of an application talk to each other. We also identify which components are vital for the system to work properly.

Our E-Commerce application has connected components.

These include:

- * Frontend services
- * Backend APIs
- * Database systems
- * Storage services
- * Monitoring systems

The application components and services are connected to each other and to database systems, storage services.

The E-Commerce application also uses monitoring systems.

Application dependency mapping helps to understand these connections.

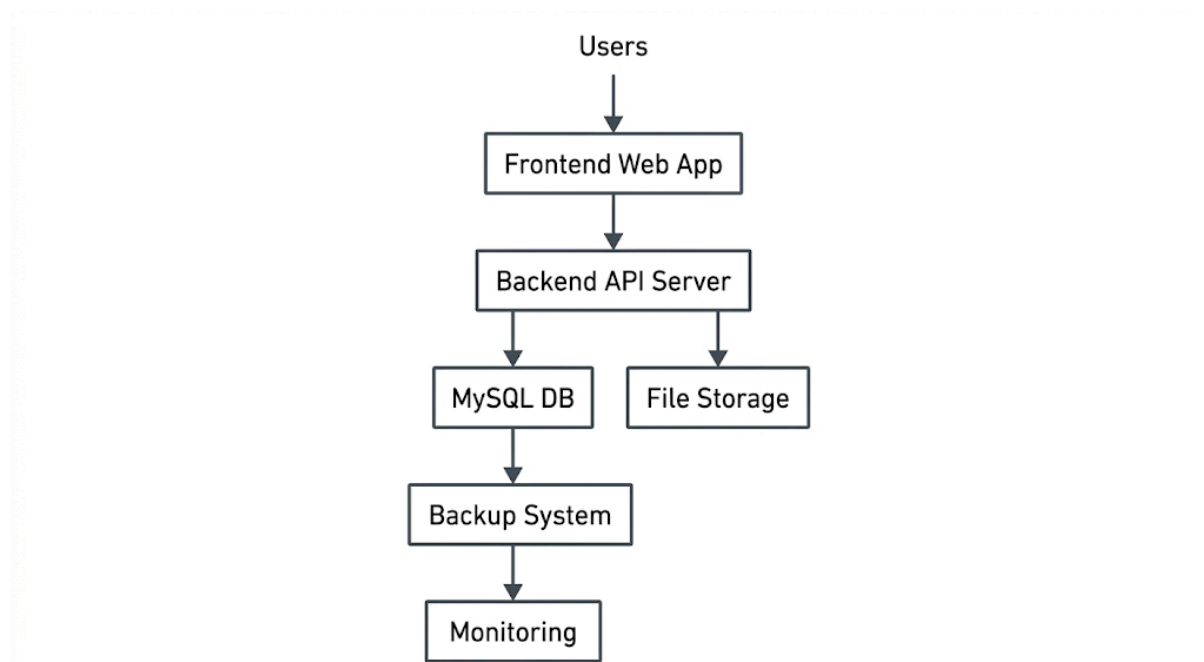
We need to map the application dependencies, for our E-Commerce application.

Application Component Dependencies

Component	Depends On	Purpose
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Frontend Web Application	Backend API	Sends user requests and displays responses
Backend API	Database Server	Processes business logic and stores data
Backend API	File Storage	Accesses product images and files
Database Server	Backup Storage	Stores backup data
Monitoring System	Application Logs	Tracks system performance and errors

Application Dependency Flow



Dependency Analysis

The Backend API Server is really important to the application architecture. Most of the application services need the Backend API Server to work properly. They depend on the Backend API Server for things like data processing and communication.

The database server and the file storage system are very important too. We need to make sure they are always available and that we have backups in case something goes wrong when we move them.

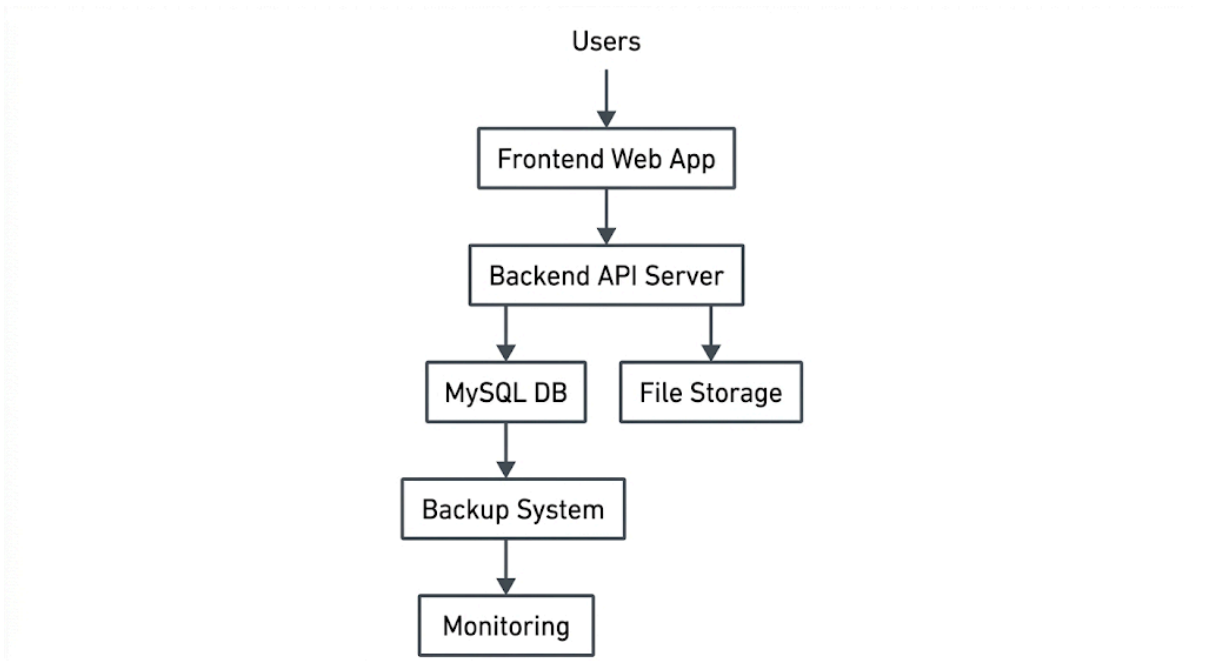
The monitoring system is not great now. It only looks at logs, from the area and it does not have a central way to monitor everything. If we use cloud-based monitoring services after we

move it will be easier to see what is going on. It will make things run more smoothly. The Backend API Server and the other services will be easier to monitor.

Migration Dependency Considerations

Dependency Area	Migration Consideration
Database Dependency	Database migration must be completed carefully to avoid data loss
Storage Dependency	File storage should be migrated to scalable cloud storage
Application Dependency	Backend services should remain available during migration
Monitoring Dependency	Centralized cloud monitoring should be implemented

Dependency Mapping Diagram (Cloud Perspective)



PROPOSED AWS CLOUD ARCHITECTURE

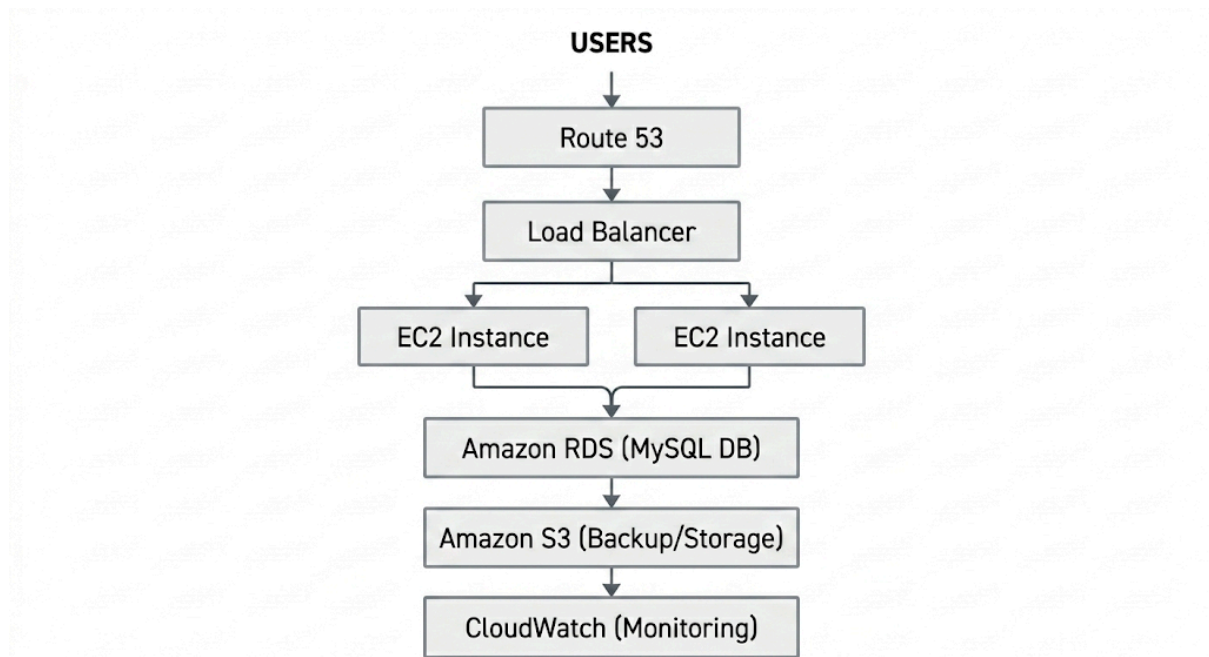
We looked at the readiness assessment and gap analysis for the E-Commerce Web Application. The result is that AWS cloud services are a choice for moving the application to the cloud. The new architecture is made to be available and scalable. It is also designed to be secure and have a central monitoring system. We will have a backup management system and a plan for disaster recovery. We are using AWS services that are managed by them.

This will reduce the work we have to do to keep everything running and make the infrastructure work better. The E-Commerce Web Application will be on the AWS cloud services. This will make things easier, for us. Improve the overall efficiency of the infrastructure.

Recommended AWS Services :

Requirement	AWS Service
Compute Services	Amazon EC2
Load Balancing	Elastic Load Balancer
Database	Amazon RDS (MySQL)
File Storage	Amazon S3
Monitoring	Amazon CloudWatch
Security Management	AWS IAM
Backup & Recovery	AWS Backup
DNS Management	Amazon Route 53

Proposed AWS Architecture Diagram



The cloud architecture that is being proposed is going to use Amazon EC2 instances to host the application servers. This means that multiple Amazon EC2 instances will be placed behind an Elastic Load Balancer. The reason for this is to make sure that traffic is distributed evenly and that the application is always available.

We have chosen Amazon RDS for the database. Amazon RDS is a managed MySQL database service. This is really helpful because it reduces the amount of work that needs to be done to administer the database. It also makes the Amazon RDS database more reliable.

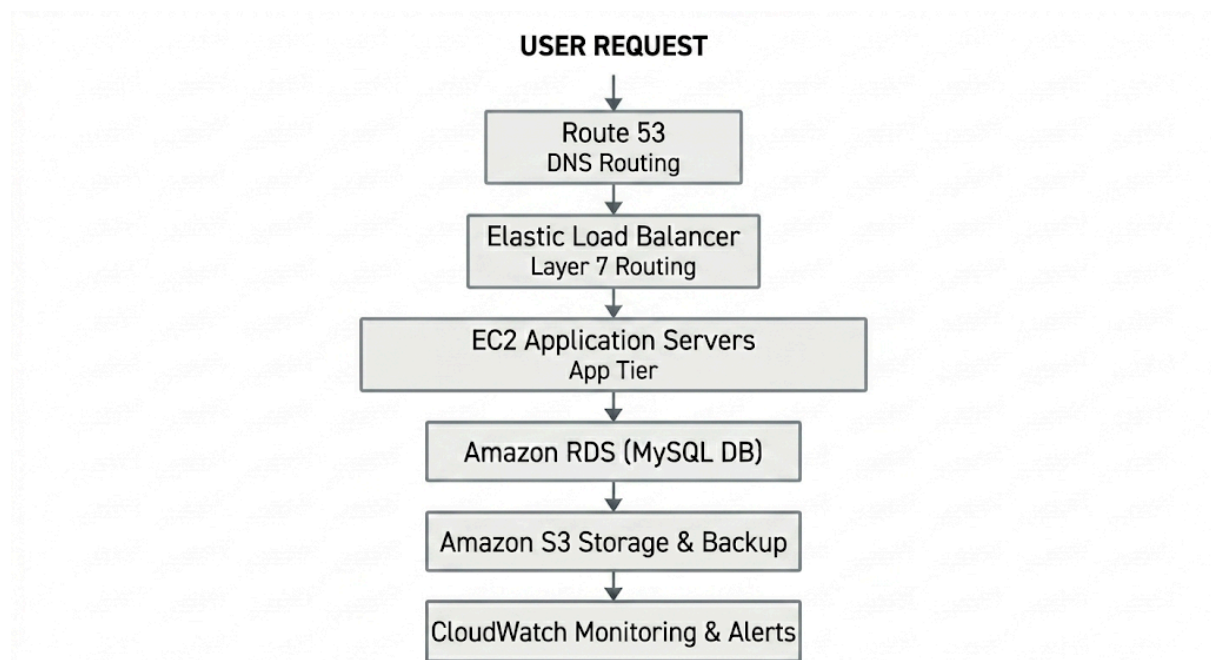
We are using Amazon S3 to store files for the application well as backups and static content. Amazon CloudWatch is what we use to monitor the infrastructure and the application. It provides a location for monitoring and logging and alerting.

Amazon EC2 instances and other cloud resources need to be secure. That is why we are using AWS IAM. AWS IAM allows us to implement access control policies that're secure and to use role-based authentication for cloud resources like Amazon EC2 instances. This helps to keep Amazon EC2 instances and other cloud resources safe, from harm. We want to make sure Amazon EC2 instances are secure.

Key Benefits of Proposed Architecture

Feature	Benefit
Load Balancer	Improved availability and traffic distribution
Multiple EC2 Instances	Better scalability and fault tolerance
Amazon RDS	Managed and reliable database service
Amazon S3	Scalable and durable storage
CloudWatch	Centralized monitoring and alerting
IAM	Improved security and access management

Cloud Architecture Workflow



TCO VS ROI COMPARISON

TCO (Total Cost of Ownership) is the total operational and infrastructure cost required to maintain the existing on-premise environment.

ROI (Return on Investment) represents the financial and operational benefits achieved after migrating the infrastructure to the cloud platform.

The comparison helps evaluate whether cloud migration is financially beneficial for the organization.

Existing On-Premise Infrastructure Cost (TCO)

Expense Type	Estimated Annual Cost
Physical Servers	₹4,00,000
Hardware Maintenance	₹1,20,000
Electricity & Cooling	₹80,000
Backup Management	₹60,000
IT Support Staff	₹2,00,000
Software Licensing	₹1,00,000
Total Estimated Cost	₹9,60,000

Estimated AWS Cloud Infrastructure Cost

AWS Service	Estimated Annual Cost
Amazon EC2	₹2,40,000
Amazon RDS	₹1,20,000
Amazon S3	₹40,000

Elastic Load Balancer	₹30,000
CloudWatch Monitoring	₹25,000
AWS Backup Services	₹45,000
Support & Maintenance	₹1,00,000
Total Estimated Cost	₹6,00,000

TCO Comparison Table

Parameter	On-Premise Infrastructure	AWS Cloud Infrastructure
Hardware Cost	High	Low
Scalability	Limited	High
Backup Management	Manual	Automated
Maintenance Effort	High	Reduced
Disaster Recovery	Complex	Simplified
Monitoring	Limited	Centralized
Security Features	Basic	Advanced
Availability	Moderate	High

ROI Analysis

Cloud Benefit	Business Impact
Reduced Infrastructure Cost	Lower operational expenditure
Improved Scalability	Better handling of peak traffic
Automated Backup	Reduced data loss risk

Centralized Monitoring	Faster issue detection
Better Security Controls	Improved protection against threats
High Availability	Improved customer experience

The proposed AWS cloud migration can significantly reduce infrastructure and maintenance expenses compared to the existing on-premise setup.

The estimated annual operational cost is expected to reduce from approximately ₹9.6 Lakhs to ₹6 Lakhs after migration to AWS cloud services.

In addition to financial savings, the organization will also benefit from improved scalability, better monitoring, automated backup management, enhanced security, and higher application availability.

ROI Percentage Estimation

Metric	Value
Current Annual Cost	₹9.6 Lakhs
Estimated Cloud Cost	₹6 Lakhs
Estimated Savings	₹3.6 Lakhs
Approximate Savings Percentage	35%–40%

CLOUD MATURITY SCORE AND ASSESSMENT MATRIX

The company checks how ready it is for the cloud by doing a maturity assessment. This assessment looks at how the organization can handle moving to the cloud. The cloud maturity assessment looks at things like the organizations infrastructure and security to see how ready they are for the cloud. The maturity score is figured out by looking at things like how safe the cloud is and how well it runs. This helps find the places that need to be improved before the company can fully move to the cloud. The current E-Commerce infrastructure was checked in areas to see if it is ready for the cloud. These areas include how well it can scale up if it can run automatically if it can be monitored, how safe it is, if the data is backed up and if the infrastructure is modern. The cloud maturity assessment checks all these things to see how ready the E-Commerce infrastructure is, for the cloud.

Cloud Maturity Assessment Table

Assessment Category	Score (Out of 5)	Remarks
Infrastructure Readiness	3	Existing infrastructure supports partial cloud migration
Security Implementation	2	Additional security controls required
Automation Capability	1	Most processes are manually managed
Monitoring & Logging	2	Centralized monitoring not implemented
Scalability Support	3	Moderate scalability support available
Backup & Disaster Recovery	2	Backup exists but lacks automation
Cloud Governance	2	Governance policies need improvement

Overall Maturity Score Calculation

Category	Score
Infrastructure	3
Security	2
Automation	1
Monitoring	2
Scalability	3
Backup & Recovery	2
Governance	2

Total Score = 15 / 35

Cloud Maturity Levels

Score Range	Maturity Level
0 – 10	Beginner Level
11 – 20	Intermediate Level
21 – 30	Advanced Level
31 – 35	Cloud Optimized Level

Based on the assessment results, the organization falls under the Intermediate Cloud Maturity Level.

The organization has basic infrastructure readiness and moderate scalability support, but improvements are required in security implementation, infrastructure automation, centralized monitoring, and disaster recovery planning.

By adopting AWS managed services and cloud best practices, the organization can improve its cloud maturity and operational efficiency.

Assessment Observations

Observation	Impact
Limited automation	Increased manual effort
Weak monitoring	Delayed issue detection
Moderate scalability	Limited traffic handling capability
Basic backup system	Increased recovery time
Security gaps	Increased security risk

Improvement Recommendations

Area	Recommendation
Automation	Implement CI/CD and infrastructure automation
Security	Enable IAM, MFA, and encryption
Monitoring	Use Amazon CloudWatch
Backup	Configure automated AWS Backup
Scalability	Implement Auto Scaling Groups

MIGRATION PLAN AND MIGRATION STRATEGY

A migration plan is a step-by-step guide that helps move all existing on-premise infrastructure, applications, databases and services to the cloud platform and in an organized way.

The goal of this migration plan is to move everything to the cloud with small downtime as possible reduce risks keep the business running smoothly and make sure the move to the cloud is successful.

Based on how ready the organization's, for the cloud and an analysis of its current infrastructure it is recommended to move to AWS cloud in phases.

Recommended Migration Strategy

Migration Strategy	Description
Rehost (Lift & Shift)	Existing application servers will be migrated to Amazon EC2 with minimal modifications
Replatform	Database and storage services will be optimized using AWS managed services
Optimization	Monitoring, backup, and security services will be improved after migration

The organization is going to move the existing application infrastructure to the AWS cloud. They will do this by using an approach that does not change the application much. This way they can avoid a lot of problems. The application will not be down for a long time.

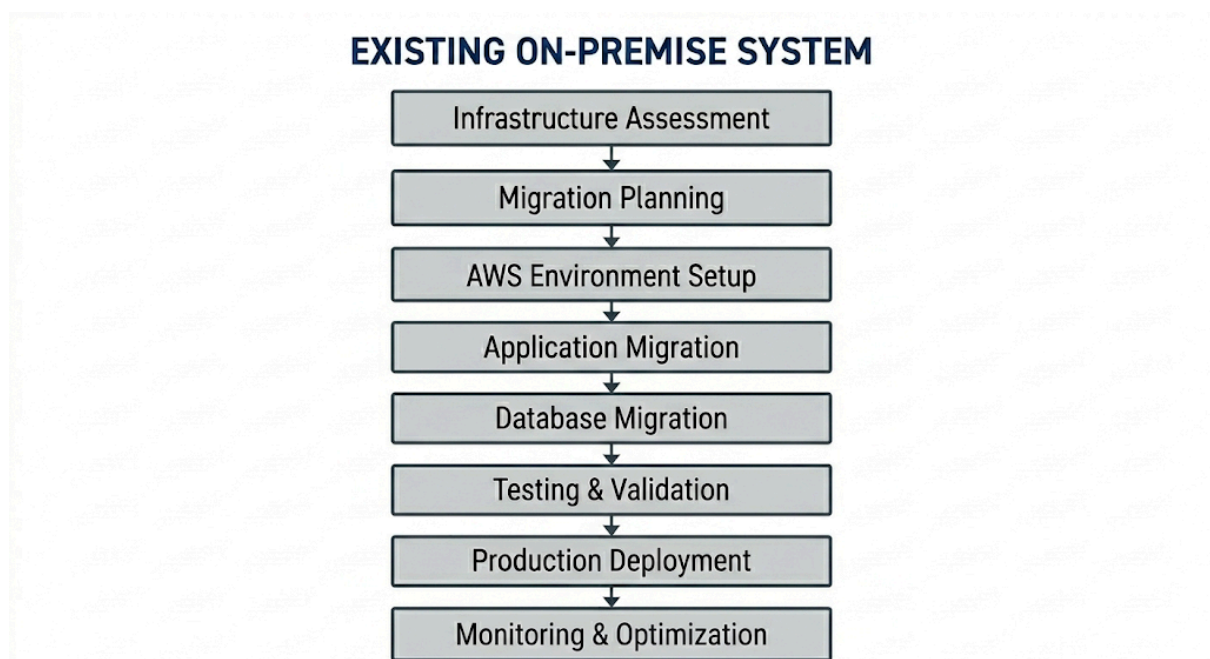
After they have moved everything they will make some parts of the infrastructure work better. These parts include the database systems, the monitoring services, the backup systems and the storage services. They will use the AWS cloud services to do this.

The organization will do the move, in steps. They want to make sure that the application keeps working and the business does not get interrupted. The AWS cloud migration will be done in a way that everything stays stable and the organization can keep working as usual.

Migration Phases

Phase	Activities
Phase 1 – Assessment	Analyze infrastructure, dependencies, and migration requirements
Phase 2 – Planning	Design AWS architecture and migration roadmap
Phase 3 – Infrastructure Setup	Configure AWS networking, IAM, and security policies
Phase 4 – Application Migration	Migrate application servers and services
Phase 5 – Database Migration	Migrate MySQL database to Amazon RDS
Phase 6 – Testing & Validation	Perform application, security, and performance testing
Phase 7 – Optimization	Configure monitoring, scaling, and backup automation

Migration Workflow Diagram



AWS Migration Components

Infrastructure Area	AWS Service
Compute Services	Amazon EC2
Database Migration	Amazon RDS
Storage Migration	Amazon S3
Monitoring	Amazon CloudWatch
Security	AWS IAM & Security Groups
Backup	AWS Backup
DNS Management	Amazon Route 53

Risk Management During Migration

Potential Risk	Mitigation Strategy
Application Downtime	Perform phased migration during low traffic
Data Loss Risk	Create full backup before migration
Security Misconfiguration	Apply IAM and security best practices
Performance Issues	Conduct performance testing after migration
Migration Failure	Maintain rollback and recovery plan

Migration Benefits

Benefit	Expected Outcome
Improved Scalability	Better handling of increased traffic
Reduced Operational Cost	Lower infrastructure maintenance cost

Better Availability	Reduced downtime
Enhanced Security	Stronger cloud security controls
Automated Backup	Improved disaster recovery capability
Centralized Monitoring	Faster issue detection and response

SECURITY RECOMMENDATIONS AND BEST PRACTICES

Security is really important when we move to the cloud. We looked at our setup and found some security problems. These problems are things like who can access our system we do not have monitoring we do not have encryption and we do security management by hand.

To make our system more secure we think we should use security services from Amazon Web Services and follow habits when we move to the cloud. We will use these security services from Amazon Web Services to fix the security problems and make our security better.

Our plan to move to the cloud will include these security services and good habits. Security is very important to us when we move to the cloud. We want to make sure we do it correctly.

We will look at our security. Make changes if we need to. Our goal is to have a system in the cloud.

Recommended Security Controls

Security Area	Recommended Solution
Identity & Access Management	Implement AWS IAM roles and policies
Authentication Security	Enable Multi-Factor Authentication (MFA)
Network Security	Configure Security Groups and NACLs
Data Protection	Enable encryption for data at rest and in transit
Monitoring & Logging	Use Amazon CloudWatch and CloudTrail
Backup Security	Implement automated AWS Backup policies
Vulnerability Management	Regular patch management and updates
Disaster Recovery	Configure backup replication and recovery strategy

AWS Security Services

AWS Service	Purpose
AWS IAM	User and role-based access management
AWS CloudTrail	Activity logging and auditing
AWS CloudWatch	Monitoring and alert management
AWS KMS	Encryption key management
AWS Shield	Protection against DDoS attacks
AWS Backup	Automated backup management
Security Groups	Network-level access control

Security Best Practices

Best Practice	Purpose
Least Privilege Access	Restrict unnecessary permissions
Enable MFA	Improve authentication security
Encrypt Sensitive Data	Protect customer information
Regular Security Audits	Identify vulnerabilities
Centralized Monitoring	Faster threat detection
Automated Backups	Improve disaster recovery
Patch Management	Reduce security vulnerabilities

The cloud system we are thinking of has ways to keep customer and business information safe. We will use Amazon RDS and Amazon S3 to store data. This data will be locked with a key using AWS Key Management Service. When people send information to the application servers it will be sent in a way using HTTPS. We will also make sure to save copies of our

data every day and keep track of everything that happens in one place. This will help us get back to normal quickly if something bad happens and it will also help us keep an eye on the security of our cloud system. The cloud system and the data stored in Amazon RDS and Amazon S3 are very important to us so we will use encryption mechanisms to protect the cloud system and the data.

Security Improvement Summary

Existing Issue	Cloud Security Improvement
Weak authentication	MFA-enabled IAM
Manual monitoring	CloudWatch monitoring
No centralized logs	CloudTrail logging
Limited backup security	Automated encrypted backups
No encryption	AWS KMS encryption

The implementation of AWS security services and cloud security best practices can significantly improve the organization's security posture.

The proposed security architecture provides stronger identity management, centralized monitoring, encrypted storage, automated backup protection, and improved threat detection capabilities.

By implementing these recommendations, the organization can reduce security risks and improve compliance with modern cloud security standards.

BENEFITS OF CLOUD MIGRATION

Cloud migration has benefits for an organization. These benefits include operational, financial and security advantages.

Moving our E-Commerce application from our on-premise setup, to AWS cloud services can help us in several ways.

- It can make our system more scalable.
- It can also make our system more available.
- This move can improve how efficiently we operate.
- It can even help ensure our business can keep running
- The cloud environment we are proposing also has advantages.
- It can make managing our infrastructure less complicated.
- This setup can also support our business as it grows in the future.

Benefits of Cloud Migration

Benefit Area	Expected Improvement
Scalability	Automatic resource scaling during high traffic
High Availability	Reduced downtime and improved reliability
Cost Optimization	Lower hardware and maintenance expenses
Security	Stronger authentication and data protection
Backup & Recovery	Automated backup and disaster recovery
Monitoring	Centralized monitoring and alert management
Performance	Faster application response and resource utilization
Operational Efficiency	Reduced manual infrastructure management

Business Benefits

Business Benefit	Impact
Improved Customer Experience	Better application performance and availability
Faster Deployment	Reduced deployment and maintenance time
Business Continuity	Improved disaster recovery support
Infrastructure Flexibility	Easy infrastructure expansion
Reduced Operational Cost	Improved return on investment

Technical Benefits

Technical Improvement	Result
Load Balanced Architecture	Better traffic handling
Managed Database Services	Reduced database maintenance effort
Cloud Storage	Scalable and durable storage
Automated Monitoring	Faster issue detection
Infrastructure Automation	Reduced manual operational tasks

This project successfully performed a cloud readiness and migration assessment for the existing E-Commerce Web Application infrastructure.

The assessment identified several technical, operational, and security-related gaps in the current on-premise environment. The proposed AWS cloud architecture addresses these limitations by introducing scalable compute resources, managed database services, centralized monitoring, automated backups, and improved security controls.

The migration strategy recommends a phased cloud adoption approach using AWS services such as Amazon EC2, Amazon RDS, Amazon S3, CloudWatch, and IAM.

The proposed solution can significantly improve application scalability, reduce operational expenses, enhance security, and improve overall system reliability.

Conclusion

The cloud readiness assessment showed that our current E-Commerce infrastructure is somewhat ready for cloud migration. We need to make some improvements. We need to improve automation, monitoring, security and disaster recovery planning.

AWS cloud services are an option for updating our existing infrastructure. They are reliable. Can handle a lot of work. This means we can reduce the complexity of our operations and the costs of maintaining our infrastructure.

The plan for migrating to the cloud and the cloud architecture we propose can help our organization in ways. We can scale better our customers will have an experience our security will be stronger and our business will be more continuous.

By using AWS cloud technologies and following security practices our organization can successfully move from a traditional infrastructure to a modern cloud-enabled environment. We will be using AWS cloud services to make this happen. This will be a change, for our E-Commerce infrastructure.